

Problem C

Smart Care Center IV

Time limit: 3 seconds

Memory limit: 1024 megabytes

Problem Description

As society ages, ensuring the safety of elderly people has become a critical issue. Falls are one of the most common and dangerous accidents in care centers.

Jim, a thoughtful developer, decides to upgrade the simulation system to Version 4.0. The care center is now a 3D multi-story building, and the system must calculate a Safety Rate to evaluate the effectiveness of the warning mechanism.

The environment is represented as a 3D Grid (Layers, Rows, Columns). '0' represents a Safe Zone, and '1' represents a Puddle (Danger).

You are given the 3D map size, the layout of the grid, and a list of elderly people attempting to walk through the center sequentially.

Jim updates the simulation rules for Version 4.0:

- Wall/Floor Logic (Safety Barrier): If an instruction causes the person to move outside the 3D Grid boundaries (including ceilings and floors), the wristband detects the barrier and IGNORES that instruction. The person stays in the current position. They do not fall.
- Puddle Logic (Danger): If an elderly person steps into a Puddle (1), they slip and fall immediately. The simulation for this person ends. The 3D coordinate of this Puddle is permanently recorded in the cloud database as a "Known Danger Zone".
- Global Warning (Prevention): If a future person attempts to step into a coordinate that is already marked as a "Known Danger Zone", the wristband vibrates and IGNORES the instruction. The person stays in the previous safe spot and continues to the next instruction. This event counts as a successful "Avoidance".

The Movement Rules are shown as follows:

- Directions are represented by integers: 0 (North), 1 (East), 2 (South), 3 (West).
- Instructions string consists of:
 - F: Move forward one step in the current direction.
 - R: Turn right (90 degrees clockwise).
 - L: Turn left (90 degrees counter-clockwise)
 - U: Go up one layer ($z + 1$).

- D: Go down one layer ($z - 1$).

The following is the Safety Rate Analysis:

After processing all cases, determine the Safety Rate based on interaction instances:

$$\text{Safety Rate} = \frac{\text{Successful Preventions}}{\text{Total Puddle Encounters}} \times 100\%$$

, where Total Puddle Encounters is the total count of times any person attempted to move into a Puddle (regardless of whether they fell or were saved), Successful Preventions is the count of times a person attempted to move into a Puddle but was saved by the Warning System (instruction ignored).

Note that If Total Puddle Encounters is 0, the Safety Rate is defined as 100.00%

Input Format

The input consists of multiple test cases until EOF. The first line contains three integers X, Y and Z representing the size of the care center (X is the columns (width), Y is the rows (depth) and Z is the layers (Height)). The following Each $Y \times Z$ lines which contains X integers consists of 0 or 1, representing the map of the care center. That is there are Z 2D grids with size of $X \times Y$. The following $2 \times N$ lines represent the N elderly persons. Each person contains 2 lines. The first line contains four integers X, Y, Z and D , which represent the starting position and the initial direction. And the second line contains a string of instructions.

Output Format

For each person, output 'Fall at (x, y, z)' if the person falls into a puddle, where (x, y, z) is the puddle's coordinate. Otherwise, output 'Safe at (x, y, z)', where (x, y, z) is the finish coordinate. After processing all test cases (at EOF), output the Safety Rate: $P\%$ (where P is formatted to 2 decimal places).

Technical Specification

- $1 \leq X, Y, Z \leq 50$
- $1 \leq x \leq X$
- $1 \leq y \leq Y$
- $1 \leq z \leq Z$
- The instruction length is at ost 100.

Sample Input 1

```
4 3 1
0 0 0 0
0 1 0 0
0 0 0 0
0 1 0 1
```

Sample Output 1

```
Fall at (1, 1, 0)
Safe at (0, 1, 0)
Safe at (0, 1, 0)
Safety Rate: 66.67%
```

```
F
0 1 0 1
F
0 1 0 1
F
```

Sample Input 2

```
3 3 2
0 0 1
0 0 0
0 0 0
0 0 1
0 0 0
0 0 0
0 0 0 1
FF
0 0 0 1
FFUF
1 0 1 1
F
```

Sample Output 2

```
Fall at (2, 0, 0)
Fall at (2, 0, 1)
Safe at (1, 0, 1)
Safety Rate: 50.00%
```

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